# Appendix A.

## Software Access and Architecture

Mac users can download NAQD here https://cutt.ly/unZDUMq, Windows users here https://cutt.ly/KnL9frz. To execute the application for the first time in a computer, we need to do the following

(1) extract the zip folder,

(2) open the folder starting with "NAQD-" it reads NAQD-Win and NAQD-Mac, for Windows and Mac operating systems, respectively, and

(3) double click on the file called "NAQD-Mac" or "NAQD-Win."

The first time NAQD is executed, you may be required to select menu "View" and "Force Reload" to generate all paths to this newly extracted folder. If after forcing reload, NAQD does not display as shown here https://cutt.ly/JnJO0MH, close the window, execute NAQD again; if needed, force reload once more when the software is launched.

NAQD is housed in a secure server location and all executable files are virus-free but may require administrative permission to be deployed. This is particularly true for Mac users, for they need to go to systems security and select, execute NAQD even if this software was not developed by Apple. Finally, each of these executable files for Mac and Windows operating systems contain all the source code required to modify and/or expand the application capabilities in the file called "app.R."

Note that NAQD has also been peer-reviewed by Code-Ocean, a multi-software platform that seeks to ensure quality, reproducibility, and transparency in scientific research. To view NAQD in Code Ocean follow this link https://cutt.ly/2riYDEhH.

***NAQD User Interface and Architecture***

Figure 9 shows NAQD's user interface (UI). This UI is configured by four parts. In the first two sections we load our data or load the data included in the distribution of NAQD for replication purposes. The third section includes the only required steps that are the selection of actors and codes. The fourth section is optional, but this section is where we can select the column in our datasets that will be used for comparison analyses. This is achieved with the "role" column. Populating this column will render the comparative analyses disaggregating the visuals by attributes, as well as conducting the QAP correlation analyses. In the following lines we describe these steps in more detail.

NAQD execution requires the following steps:

A. Load your data or select the example provided.
    a. NAQD describes the example that comes preloaded.
    b. When you load your database, in comma separated value (CSV) format, these data will be displayed in this main panel.
B. Data access
    a. To upload a database, select "Click to browse..." and search for a file stored locally (see Figure 10(a)).

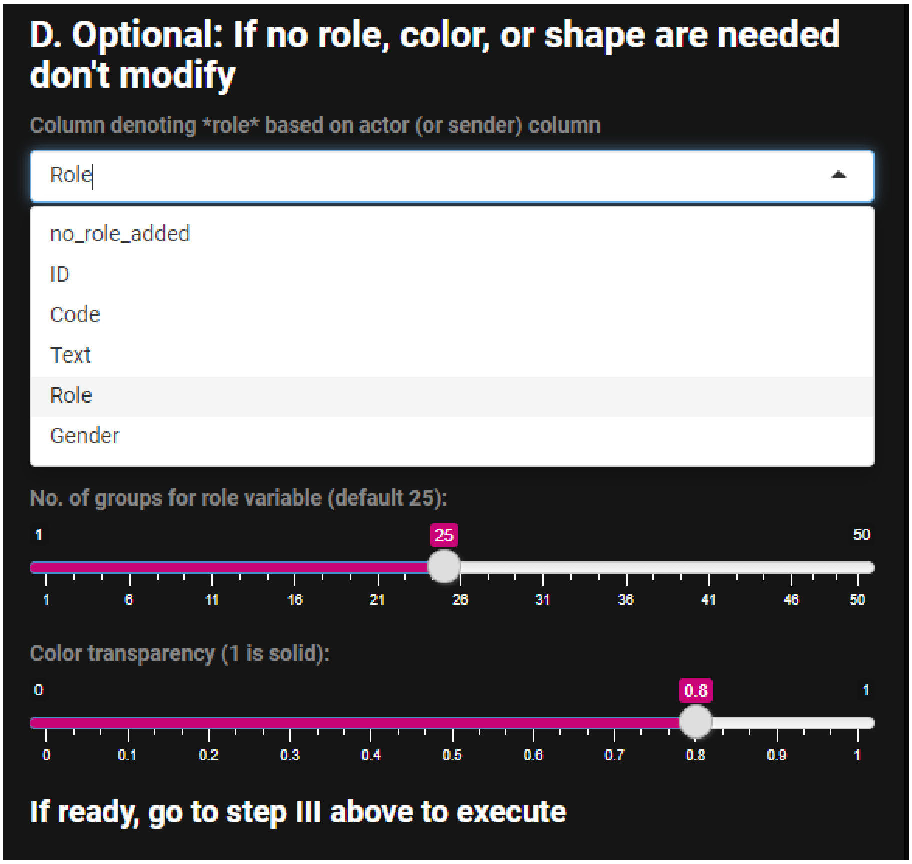
    b. Alternatively, you can load the data example provided by selecting "Interview or multiple focus groups example" under section A and selecting "Click to load data example shown below" (see Figure 10(b)).

C. Select the "actor" column (or provider of code)

D. Select the "code" column.

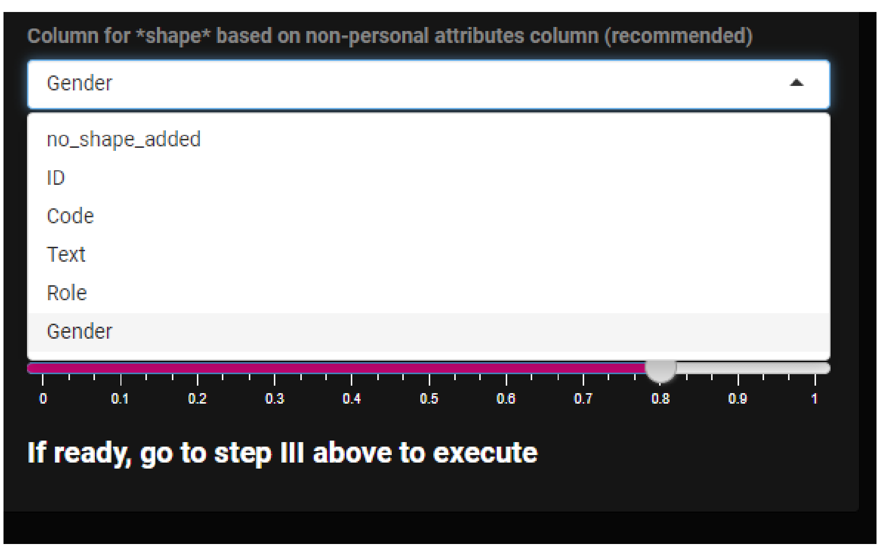
E. OPTIONAL (these pieces of information aid in the contextualization of information generation or allow the execution of QAPs):

    a. Column denoting ***role***. It indicates specific roles of the actors (e.g., students, faculty, administrators). If selected, all combinations will automatically be tested using QAPs and network visualizations will be produced. **It is recommended to have at least three actors per role to conduct QAPs**.

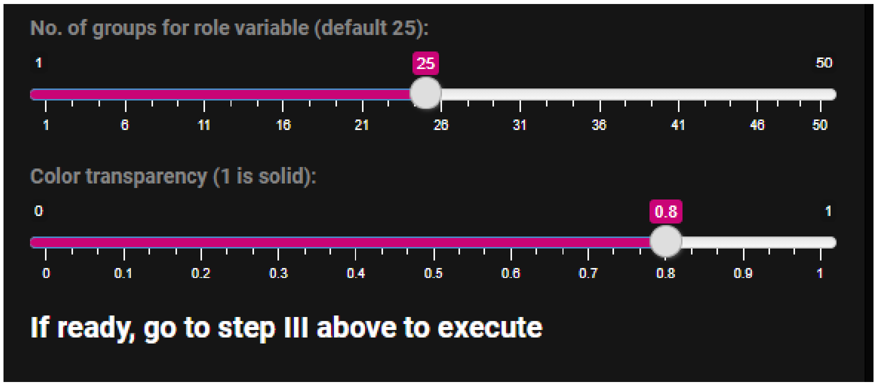



b. Column for ***color***. By default, the colors assigned are calculated based on ***community*** ascription using spin-glass modeling. All codes are orange in the visualization (this color can be changed in the optional adjustments tab).


c. Column denoting ***shape*** that may further help in the contextualization of the information provided by actors. This attribute may be personal (e.g., gender) or non-personal (e.g., field of study) in addition to be displayed when clicking on an actor, this attribute will be reflected in the shape of the nodes in the network. All codes are a circle in the visualization (this shape can be changed in the optional adjustments tab).

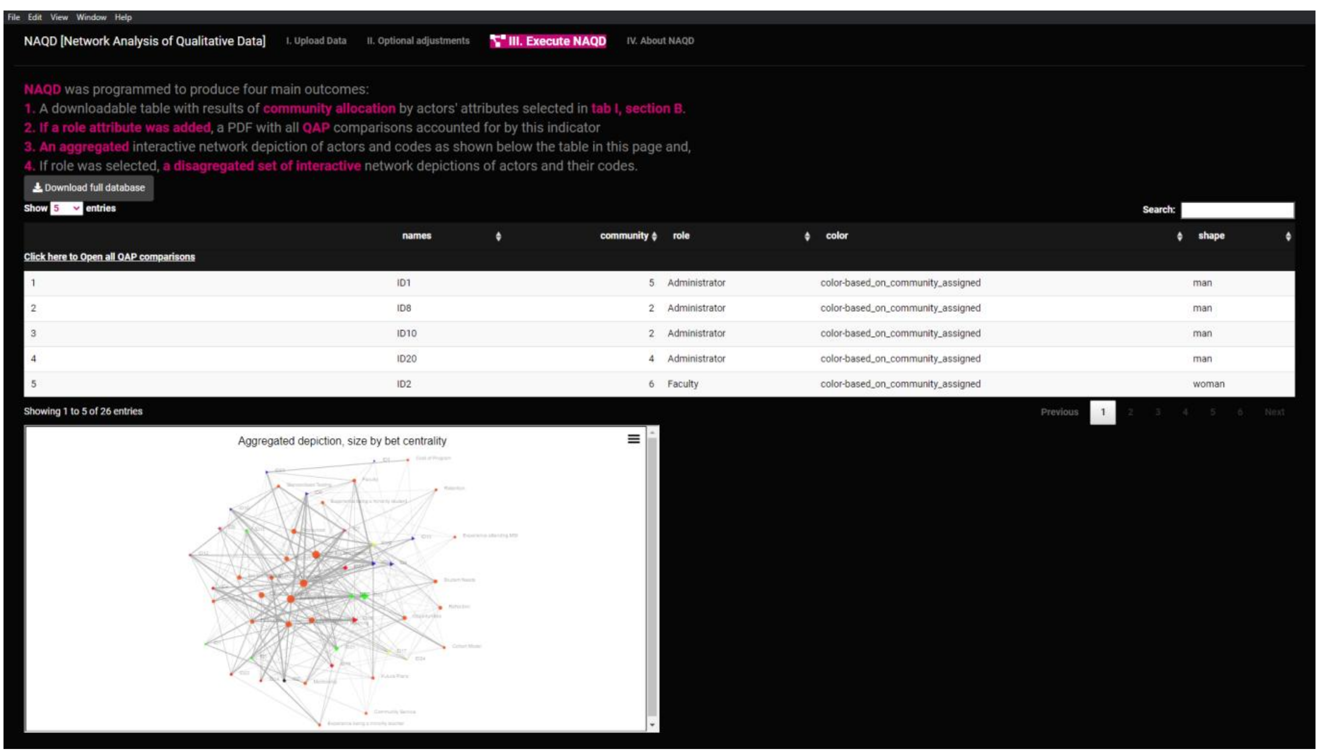


d. Researchers can modify the ***number of optimal number of communities*** to detect. The recommended number is 25 (Csardi & Nepusz, 2006). This number does not mean that 25 communities will be detected, but that the maximum may be 25.



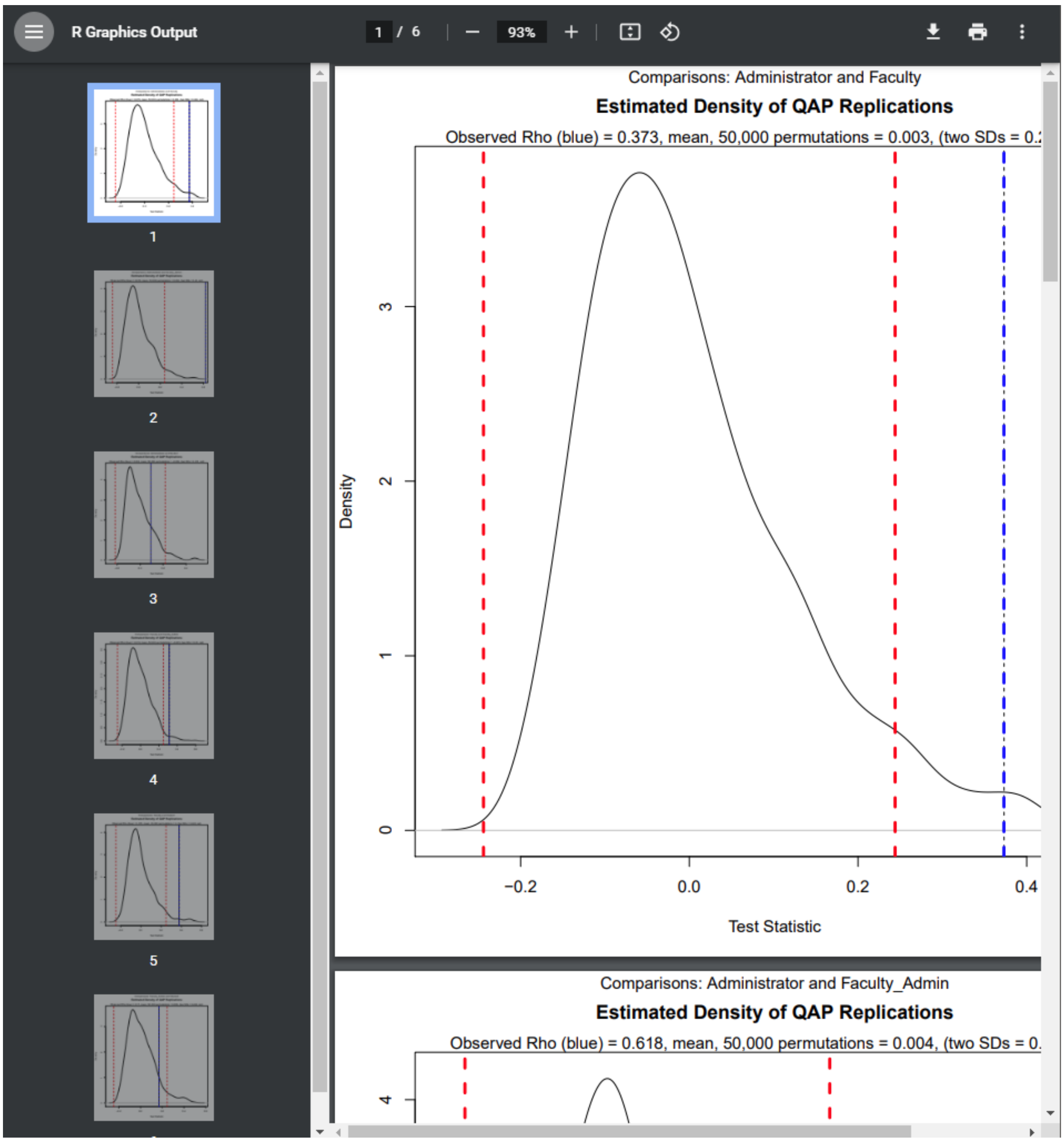

e. You can also select the ***transparency*** level of the colors to be plotted, with 1 being solid and 0 being transparent. Recommended value is 0.8.

F. When ready, "Execute NAQD" on the top panel to obtain all four outputs.

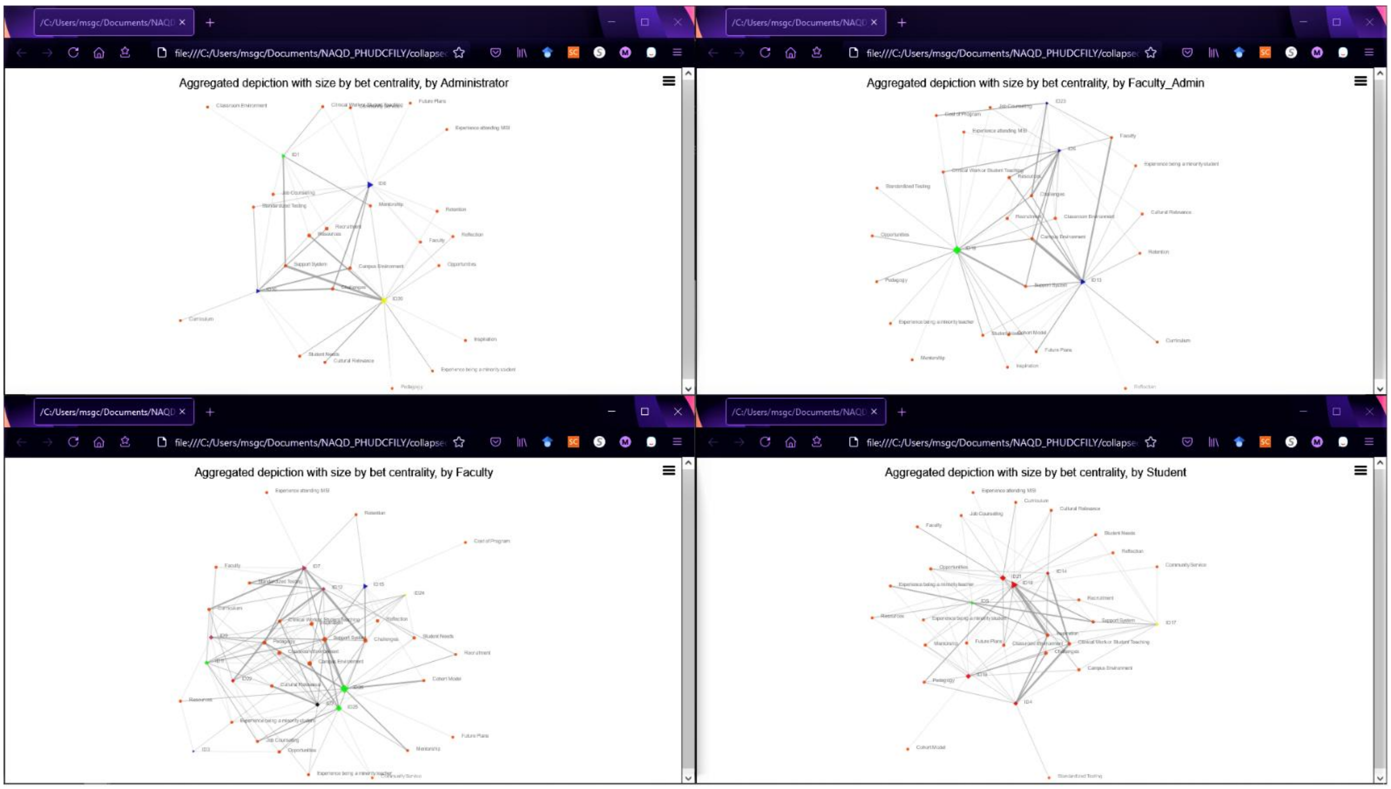


a. The database displayed may be downloaded in csv format.

b. If role was selected, a PDF will be generated containing all combinations based on roles by clicking "Click here to Open all QAP comparisons" on top of the table.



c. In addition to the aggregated network depiction shown here, this network will be automatically launched in your browser window with the address indicating the local folder where this (these) file(s) is (are) stored. Moreover, all *n* disaggregated network depictions by role will also be launched.




d. All HTML files can be easily distributed as standalone files. Their display does not need internet connection.